# Microsoft

# Data Engineer Interview Questions (0-3 years) 25 LPA

## SQL + DataModeling

1. Write a query to identify users who placed more than 2 orders in a week but skipped one specific weekday.
2. How would you handle Slowly Changing Dimensions (Type 2) in Azure Synapse?
3. What partitioning and indexing strategies would you use for a billion-row fact table?
4. Explain the difference between CUBE, ROLLUP, and GROUPING SETS.
5. How do you optimize a query that's performing poorly?
6. When would you choose star schema vs snowflake schema in data modeling?
7. What are window functions? Give an example use case with ROW_NUMBER, RANK, or LAG/LEAD.
8. How would you detect duplicate records in a table and remove them efficiently?
9. Explain normalization vs denormalization with examples.
10. How do you handle late-arriving dimensions in ETL?

## Python + DSA

1. **Given a stream of events (user_id, timestamp), detect fraud patterns based on time gaps.**
2. **Implement a string compression function that supports UTF-8 multi-byte characters.**
3. **Design an LRU cache using files instead of memory, tracking read/write costs.**
4. **Explain time complexity of different Python data structures (dict, set, list).**
5. **How do you handle memory management in Python?**
6. **Implement a function to check if a string is a valid palindrome, ignoring punctuation.**
7. **How would you implement a producer-consumer system in Python?**
8. **Solve: Find the k-th largest element in an unsorted list .**
9. **Implement a custom exception class and show how you'd use it in error handling.**
10. **Explain the difference between deep copy vs shallow copy in Python.**

## PySpark + System Design

1. Design a real-time recommendation system with <1s latency for millions of users.
2. How would you optimize joins in PySpark (broadcast vs shuffle)?
3. Explain when you would use caching vs persistence in PySpark.
4. Describe how you'd build a data pipeline with schema evolution (Bronze → Silver → Gold).
5. How do you debug and optimize a slow PySpark job?
6. What is predicate pushdown and why does it matter?
7. Explain the difference between repartition and coalesce.
8. How do you monitor PySpark jobs in production?
9. When would you use bucketing in Spark SQL?
10. Explain fault tolerance in Spark (lineage + DAG).
11. How would you design a CDC (Change Data Capture) pipeline in Spark with exactly-once guarantees?
12. Compare batch vs streaming in Spark. When would you choose each?